

Analytical Report:
Predicting and Explaining Customer Churn

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By:
Soochan Woo

Executive Summary

In this report, I use the data mining tool KNIME to build and compare predictive models for a dataset of customer data, to perform customer churn analysis. The five predictive models that were chosen are Decision Tree, Random Forest, Artificial Neural Network, Logistic Regression, and Boosted Tree. The final objectives are to compare the model performances, select the best performing model, and to make suggestions for how the best performing model can be utilized for customer churn analysis.

This report begins with a statement about the business objectives, namely to select a model for customer churn analysis in order to reduce customer churn, and an exploration of the data. The dataset itself is relatively small, with just 1,000 data points, where each data point represents a distinct customer. The report then moves on to detail the data preparation, where feature engineering, data balancing, and other data preprocessing steps are applied to prepare the data for modeling.

Using KNIME, I input the prepared data into the chosen models, and generate predictions from each node. I use Scorer and ROC Curve nodes to compare the models' performances based on accuracy, sensitivity, specificity, and ROC value, and conclude that the Random Forest model performs the best in predicting customer churn. The report concludes with suggestions to apply the Random Forest model on larger datasets of customer data to identify the customers that are likely to churn, and to deploy measures such as advertising deals to prevent high-risk customers from churning. A picture of the complete KNIME workflow is attached at the end of the report.

Business Understanding

The purpose of this report is to build and compare predictive models using a dataset of customer data to identify the best performing model for customer churn analysis. Customer churn is a significant issue for businesses due to the revenue loss that it causes, owing to factors such as customer dissatisfaction, poor customer service, and unproductive marketing. Therefore, reducing customer churn and identifying customers that are at risk of churning are crucial steps to retaining customers and preventing both immediate and long-term revenue loss for businesses.

Identifying an effective predictive model for customer churn is necessary for reducing customer churn. An effective predictive model will help to identify customers before they churn; retaining customers is almost always more cost-effective than acquiring new customers or

bringing back ones that churned. Having such a model will allow businesses to deploy marketing strategies to prevent high-risk customers from churning in the first place, such as offering deals and incentives to stay as customers.

Data Understanding:

The dataset that was chosen for modeling consists of data on 1,000 customers, with each customer corresponding to a single data point. The dataset contains 38 variables, including socio-demographic attributes such as the region where the customer lives, and behavioral attributes such as if the patient has internet access. The dataset also contains a customer ID variable for each customer and a target variable that will be used for modeling that identifies whether the customer churned or not. The data appears to be deidentified, and I have full access to the data.

Figure 1 – Statistical Overview of Variables

Name	Type	# Missing values	# Unique values	Minimum	Maximum	Mean	Mean Absolute Deviat...	Standard Deviation
Address	Number (Float)	0	46	0	55	11.571	8.127	10.128
Age	Number (Float)	0	57	18	77	41.536	10.532	12.687
Callicard	String	0	2	0	1	0	0	0
Callid	String	0	2	0	1	0	0	0
Callwait	String	0	2	0	1	0	0	0
Cardmon	Number (Float)	0	141	0	109.25	13.865	10.538	14.382
Cardten	Number (Float)	0	231	0	7,515	614.449	591.092	874.981
Confer	String	0	3	0	1	0	0	0
Custcat	String	0	4	0	1	0	0	0
Ebill	String	0	3	0	1	0	0	0
Education	String	0	5	0	1	0	0	0
Employ	Number (Float)	0	43	0	47	10.783	8.159	10.15
Equip	String	0	2	0	1	0	0	0
Equipmon	Number (Float)	0	168	0	77.7	13.599	17.115	18.86
Equipten	Number (Float)	0	185	0	4,758.05	441.176	612.963	848.871
Forward	String	0	2	0	1	0	0	0
GeFemaleder	String	0	2	0	1	0	0	0
Income	Number (Float)	0	159	9	732	71.844	48.911	80.349
Internet	String	0	2	0	1	0	0	0
InInc	Number (Float)	0	159	2.197	6.596	3.925	0.623	0.779
LogLong	Number (Float)	0	288	0.049	4.605	2.15	0.578	0.717
LongMon	Number (Float)	0	288	1.05	99.95	11.175	6.37	9.546
LongTen	Number (Float)	0	492	1.05	7,257.6	537.637	481.889	730.353
Marital	String	0	2	0	1	0	0	0
Multiline	String	0	2	0	1	0	0	0
Pager	String	0	2	0	1	0	0	0
Region	String	0	3	0	1	0	0	0
Reside	Number (Float)	0	7	1	7	2.313	1.175	1.434
Retire	String	0	2	0	1	0	0	0
Tenure	Number (Float)	0	73	1	72	35.231	18.576	21.316
Tollfree	String	0	2	0	1	0	0	0
Tollmon	Number (Float)	0	117	0	173	13.299	14.094	17.401
Tollten	Number (Float)	0	236	0	4,905.85	533.426	655.369	866.964
Voice	String	0	2	0	1	0	0	0
Wireless	String	0	2	0	1	0	0	0
Wiremon	Number (Float)	0	133	0	109.7	10.863	15.609	19.24
Wireten	Number (Float)	0	142	0	6,444.95	422.377	636.682	961.563
Churn (Y/N)	String	0	2	0	1	0	0	0

Figure 1 provides general statistics for each of our variables after some basic data preprocessing. Of relevance to the report is the number of missing values, which has been reduced to zero for all the variables.

Figure 2 – Correlation Matrix

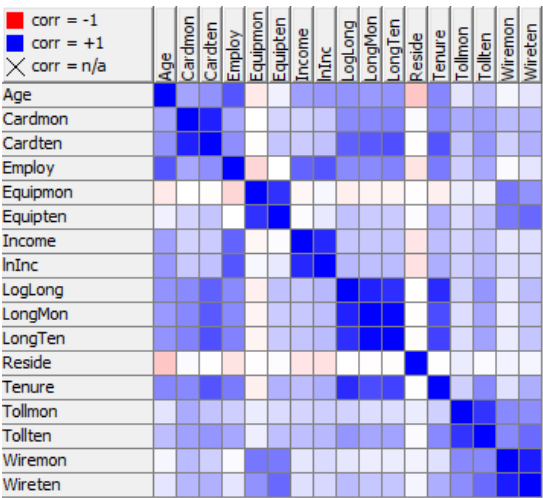


Figure 2 provides a correlation matrix of most of the numerical variables, which identifies some variable pairs that seem to have higher correlation.

Figure 3 – Frequencies of Churn (Y/N) Values

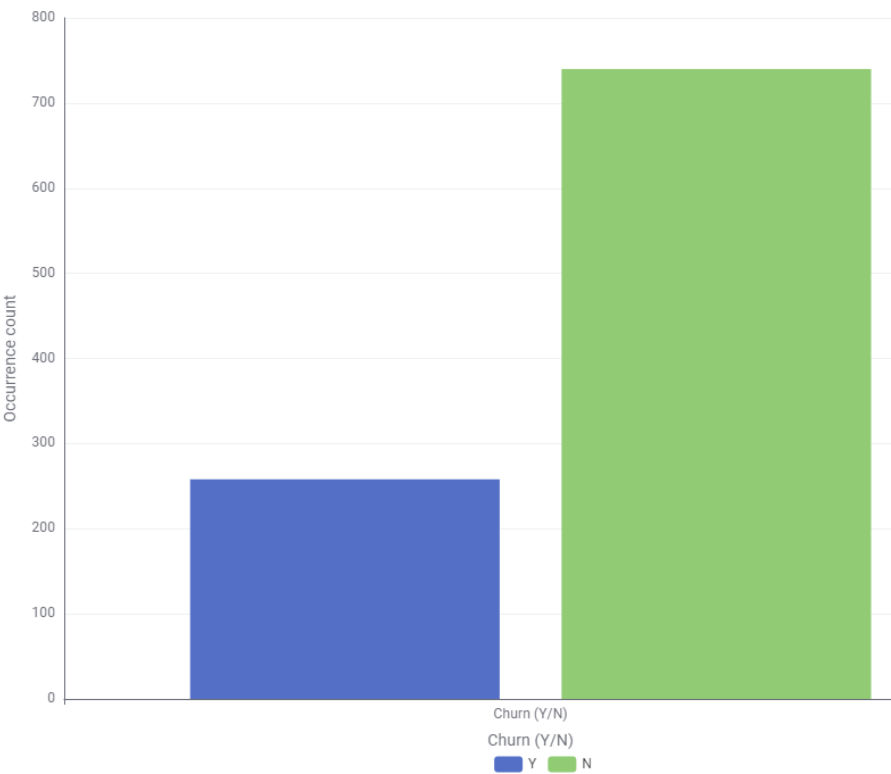


Figure 3 displays the relative frequencies of the Y (Yes) and N (No) values for whether the customer churned or not. This is helpful for modeling, since I can keep in mind that there are roughly 3 times more N values than Y values, and it may be useful for me to equalize the frequencies in my later analyses.

Data Preparation

Figure 4 – Data Preprocessing

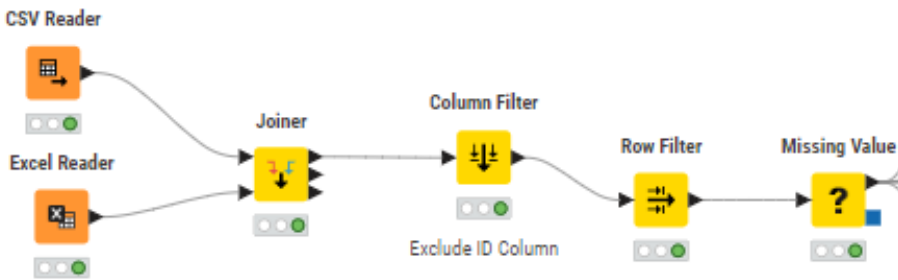


Figure 4 shows the workflow for the data preprocessing steps that were performed. The initial data was split into a CSV file containing all of the independent variables, and an Excel file containing the churn (target) variable, with both files containing the same 1,000 customer IDs. In KNIME, the Joiner node was used to join the two files on the customer ID column, to create a single table with both the independent variables and the target variable. A column filter was applied to remove the customer ID variables and the additional Description variable for the data since they would be irrelevant for the upcoming analysis. A row filter was applied to select for the data points where the churn variable did not have a missing value, resulting in the loss of just two data points. The remaining 998 data points were filtered through the Missing Value node, where integer variables, other than Age, had their missing values imputed with the column's median value, string variables had their missing values imputed with "NA", and float variables and the Age variable had their missing values imputed with the column's mean value. The resulting dataset was placed into a Color Manager node to color the "N" values of the churn variable green and the "Y" values of the churn variable red.

Modeling

After having processed the data, the data was utilized for five different predictive models: Decision Tree, Random Forest, Artificial Neural Network, Logistic Regression, and Boosted Tree. All five models used the data that had gone through the Color Manager node, and all five models were assessed by a Scorer node and a ROC Curve node.

Figure 5 – Decision Tree

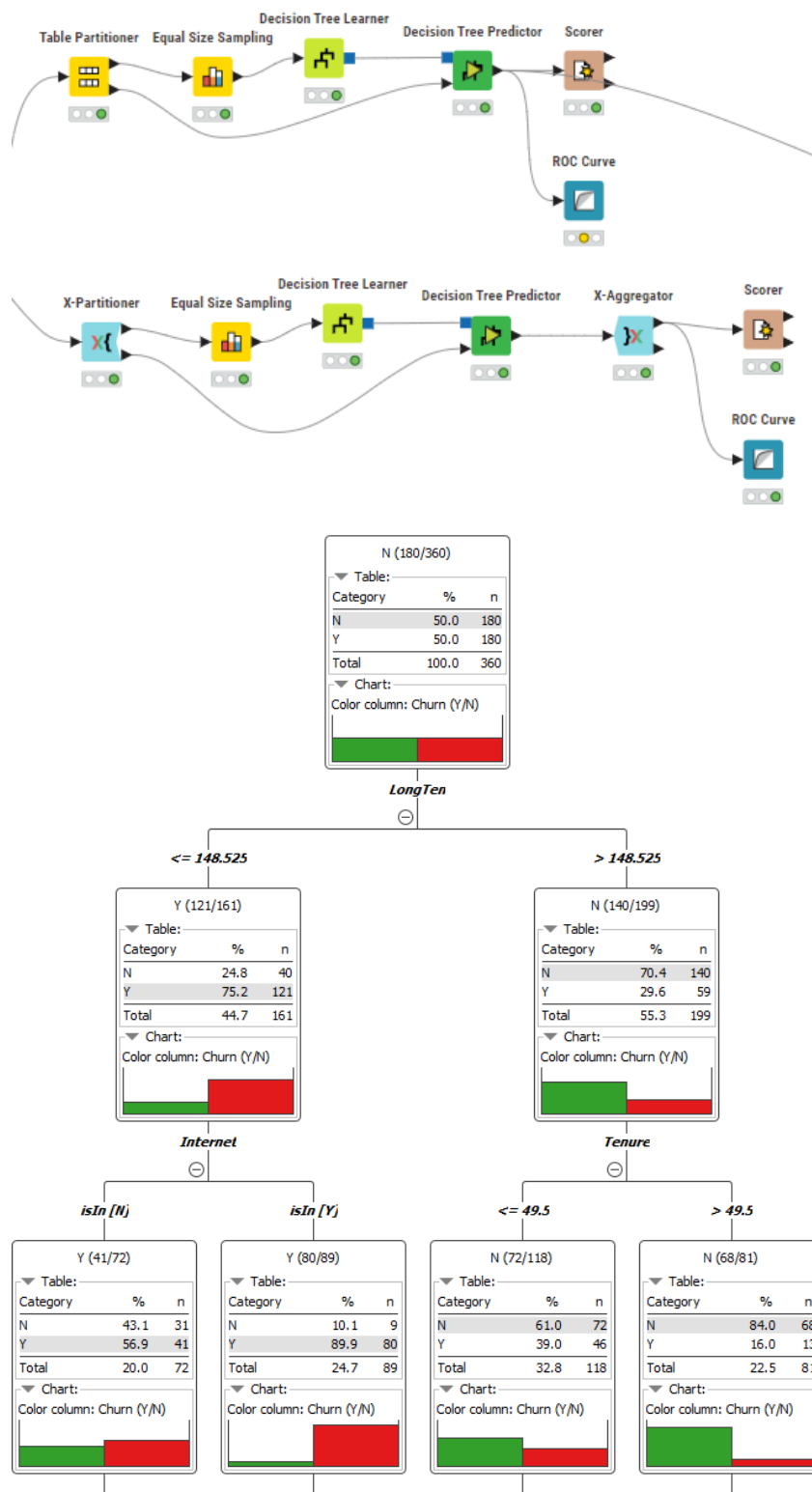


Figure 5 shows the workflow for the decision tree model, as well as the first three levels of the decision tree graphical model. As can be seen from the workflow, both single-split (top half of

workflow) and k-fold cross validation (bottom half of workflow) were applied to the model; the results of both will be compared in the following figure. The single-split was used to partition 70% of the data into a training set and 30% of the data into a test set, while k-fold cross validation was used to create 10 folds for cross validation. Equal size sampling was used to make the number of Churn=Y rows equal to Churn=N rows by taking all the Churn=Y rows and taking an equal number of Churn=N rows for analysis. The snapshot of the first three levels of the decision tree model show that the first split occurred where the LongTen (representing long tenure) variable had values less than or equal to 148.525 or greater than 148.525. The former split based on whether the Internet value was “N” or “Y” to indicate whether the customer used Internet, and the former split based on Tenure being less than or equal to 49.5 or greater than 49.5. These splits indicate that the most important variable in predicting customer churn is likely LongTen, with Internet and Tenure ranking second in predictive importance.

Figure 6 – Decision Tree: Single Split vs. K-Fold Cross Validation

Churn (Y/N...	Y	N		Churn (Y/N...	Y	N	
Y	57	21		Y	220	38	
N	56	166		N	151	589	
Correct classified: 223				Correct classified: 809			
Wrong classified: 77				Wrong classified: 189			
Accuracy: 74.333%				Accuracy: 81.062%			
Error: 25.667%				Error: 18.938%			
Cohen's kappa (κ): 0.418%				Cohen's kappa (κ): 0.568%			

Figure 6 shows the difference in accuracy between using single-split (left) and using k-fold cross validation (right). While using single-split yielded an accuracy of roughly 74.3%, using k-fold cross validation yielded an accuracy of roughly 81.1%. Due to the superior accuracy, k-fold cross validation yielded the model of choice between the two decision tree models.

Figure 7 – Decision Tree Performance Metrics and ROC Curve

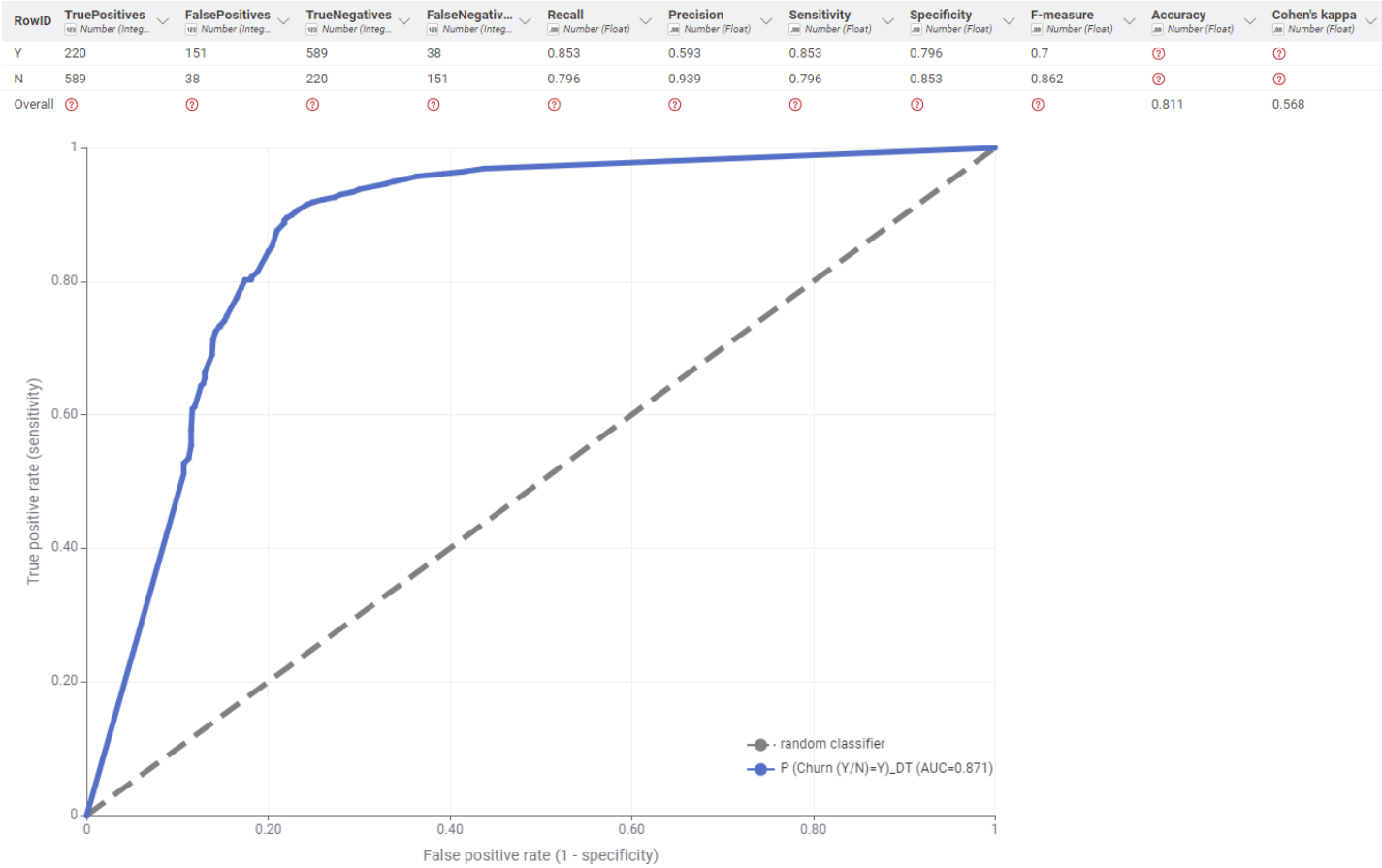


Figure 7 shows the performance metrics of the decision tree model, such as sensitivity, specificity, and accuracy, as well as the ROC curve. The sensitivity and specificity values are high, both around 0.8, and the predictive accuracy is also high at a value of 0.811, indicating that the model performs well in identifying true positives and true negatives, and in making correct predictions overall. The ROC curve shows an AUC value of 0.871, which is relatively high, indicating that the model does well in distinguishing the positive and negative outcomes.

Figure 8 – Random Forest

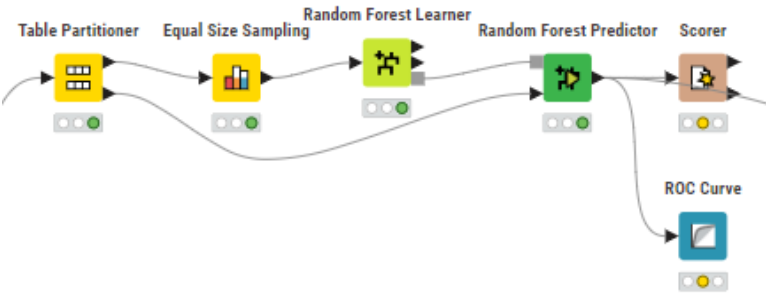


Figure 8 shows the workflow for the random forest model. Just like the decision tree model, single split partitioning was used to split the data into a training set with 70% of the data and a test set with 30% of the data. Equal size sampling was applied identically to the decision tree model. The random forest model generated and used the input of 100 decision tree models to create a more powerful predictive model, using the philosophy of ensemble learning.

Figure 9 – Random Forest Performance Metrics and ROC Curve

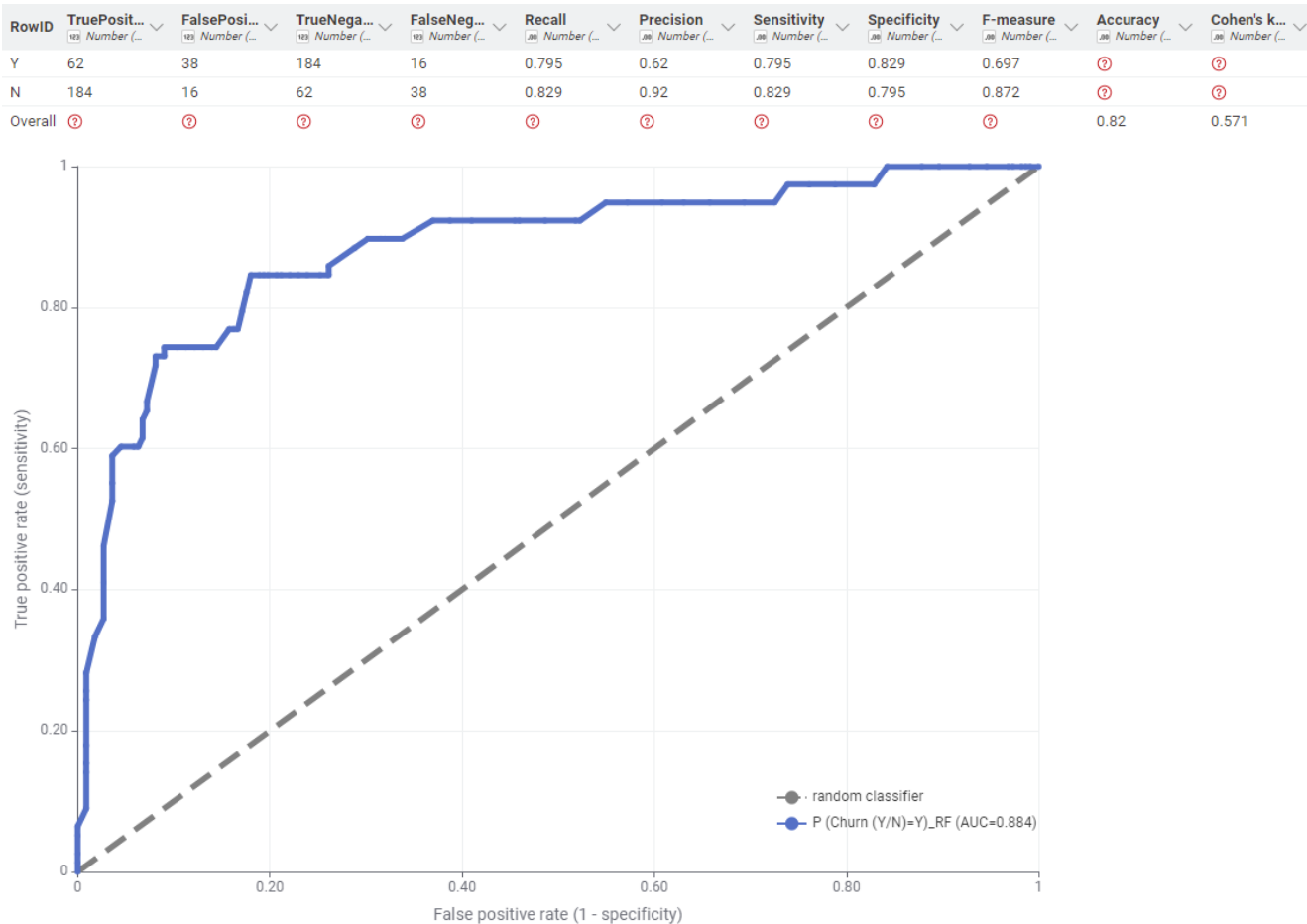


Figure 9 shows the performance metrics of the random forest model, such as sensitivity, specificity, and accuracy, as well as the ROC curve. The sensitivity and specificity values are high, both around 0.8, and the predictive accuracy is also high at a value of 0.82, indicating that the model performs well in identifying true positives and true negatives, and in making correct predictions overall. The ROC curve shows an AUC value of 0.884, which is relatively high, indicating that the model does well in distinguishing the positive and negative outcomes.

Figure 10 – Artificial Neural Network

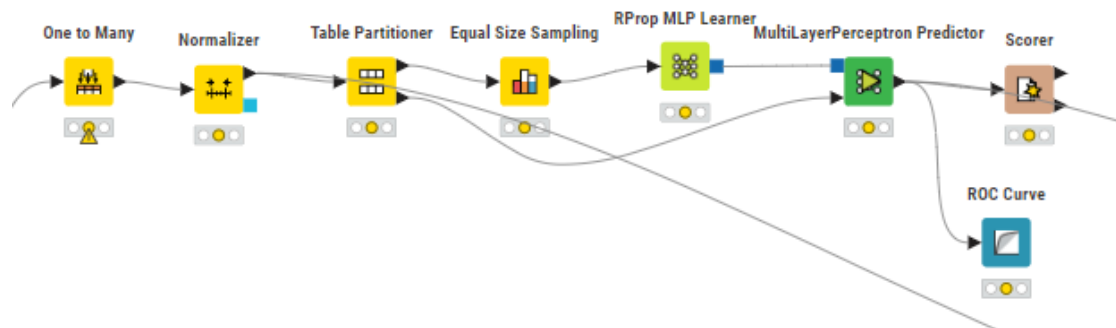


Figure 10 shows the workflow for the artificial neural network model. Unlike the set-based prediction models in Figures 5 and 8, the artificial neural network model is a number-based model, and requires all values to be numerical. Therefore, one-hot encoding was used to assign proportional numerical values to non-numerical variables, and min-max normalization was applied to contain all the values of each variable between 0 (representing the minimum) and 1 (representing the maximum). Similar to the previous two models, this model used single-split partitioning and equal size sampling with the same settings as the previous two models. The artificial neural network model imitates the thinking process of the human brain to perform predictive analysis on the data.

Figure 11 – Artificial Neural Network Performance Metrics and ROC Curve

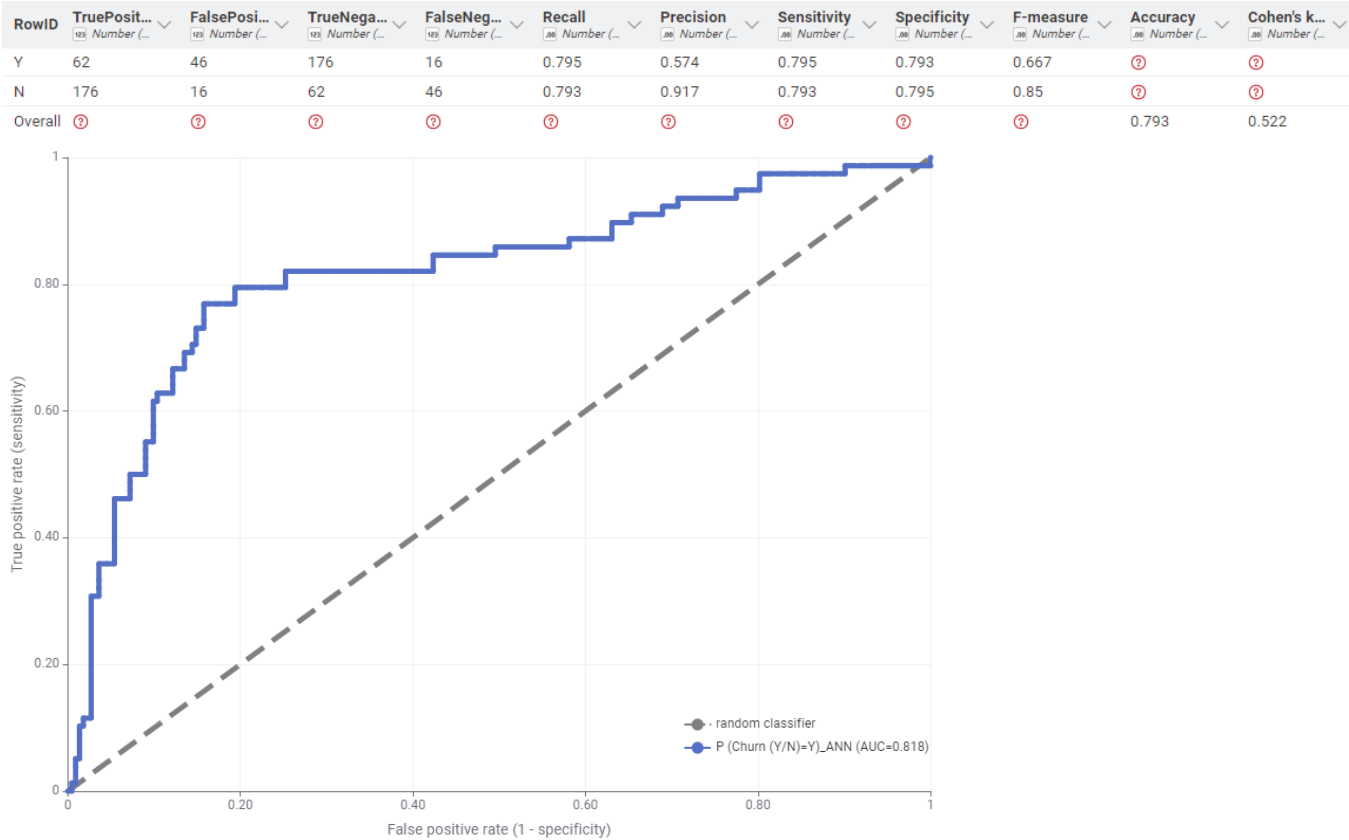


Figure 11 shows the performance metrics of the artificial neural network model, such as sensitivity, specificity, and accuracy, as well as the ROC curve. The sensitivity and specificity values are high, both slightly less than 0.8, and the predictive accuracy is also high at a value of 0.793, indicating that the model does well in identifying true positives and true negatives, and in making correct predictions. The ROC curve shows an AUC value of 0.818, which is relatively high, indicating that the model does well in distinguishing the positive and negative outcomes.

Figure 12 – Logistic Regression

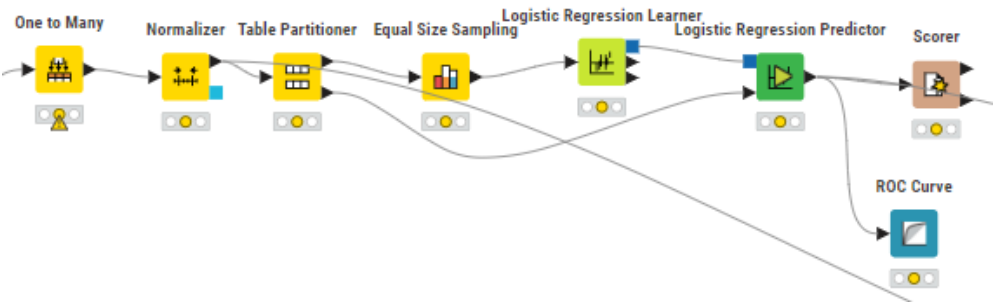


Figure 12 shows the workflow for the logistic regression model. Because it is a number-based predictive model, one-hot encoding was used to assign proportional numerical values to non-numerical variables, and min-max normalization was applied to contain all the values of each variable between 0 (representing the minimum) and 1 (representing the maximum). Single-split partitioning and equal size sampling were used with the same settings as the prior three models. The logistic regression model uses the logarithm of the odds of a customer churning to estimate the probability of customer churn.

Figure 13 – Logistic Regression Performance Metrics and ROC Curve

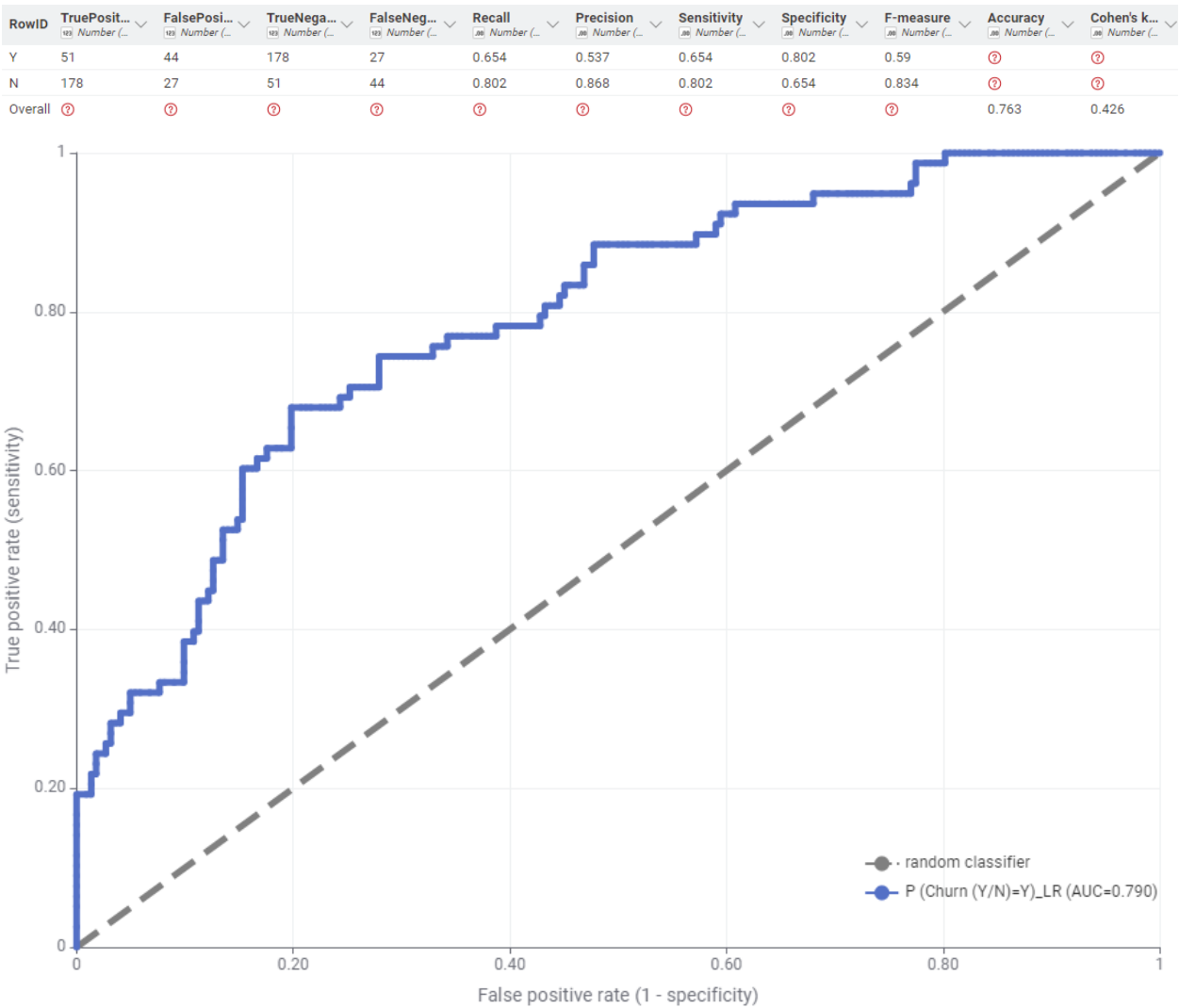


Figure 13 shows the performance metrics of the logistic regression model, such as sensitivity, specificity, and accuracy, as well as the ROC curve. The sensitivity value is somewhat low at

0.654, while the specificity value is somewhat high at 0.802, and the predictive accuracy is also high at a value of 0.793. This indicates that the model does well in identifying true negatives and in making correct predictions, but not so much at identifying true positives. The ROC curve shows an AUC value of 0.790, which is relatively high, indicating that the model does well in distinguishing the positive and negative outcomes.

Figure 14 – Boosted Tree

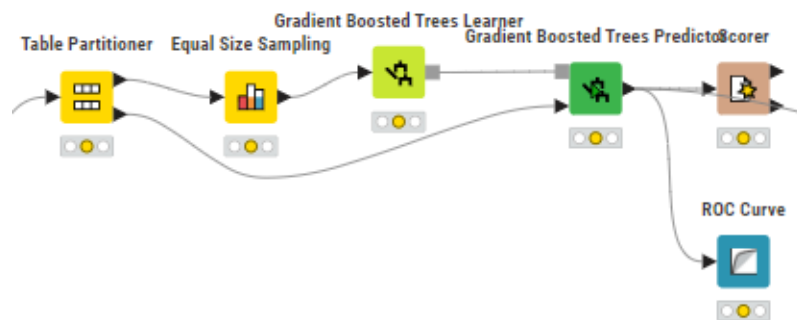


Figure 14 shows the workflow for the boosted tree model. Like the other two set-based models in Figures 5 and 8, the boosted tree model does not use one-hot encoding, and uses single-split partitioning and equal size sampling with the same settings as all the prior models. The boosted tree model is similar to the random forest model in that it builds multiple decision trees; for this model, 100 models were used. The model is unique in that each subsequent tree corrects the errors made by the previous trees, in order to sequentially improve predictive accuracy.

Figure 15 – Boosted Tree Performance Metrics and ROC Curve

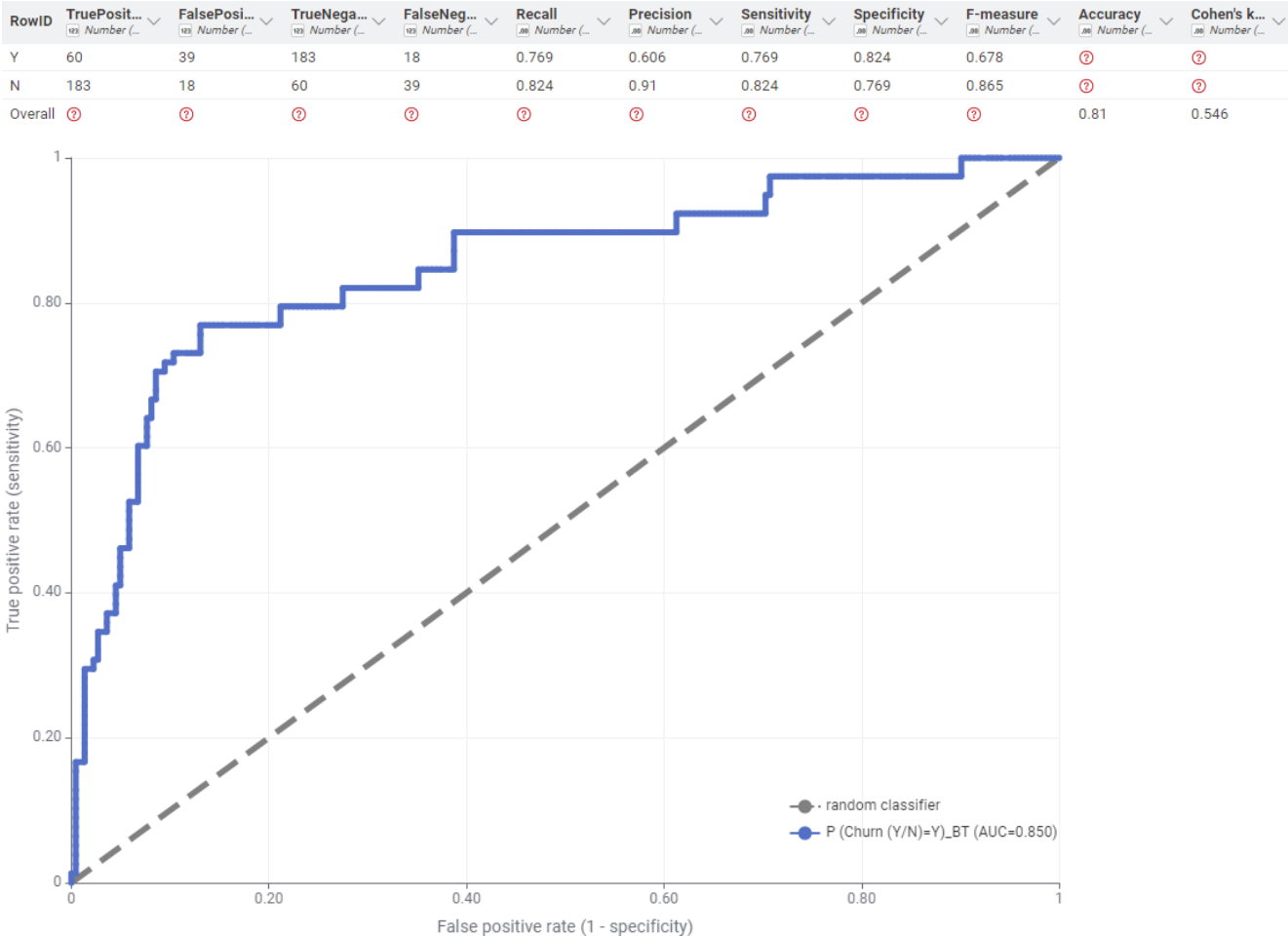


Figure 15 shows the performance metrics of the boosted tree model, such as sensitivity, specificity, and accuracy, as well as the ROC curve. The sensitivity and specificity values are high, both around 0.8, and the predictive accuracy is also high at a value of 0.81, indicating that the model does well in identifying true positives and true negatives, and in making correct predictions. The ROC curve shows an AUC value of 0.850, which is relatively high, indicating that the model does well in distinguishing the positive and negative outcomes.

Evaluation

Figure 16 – ROC Curves

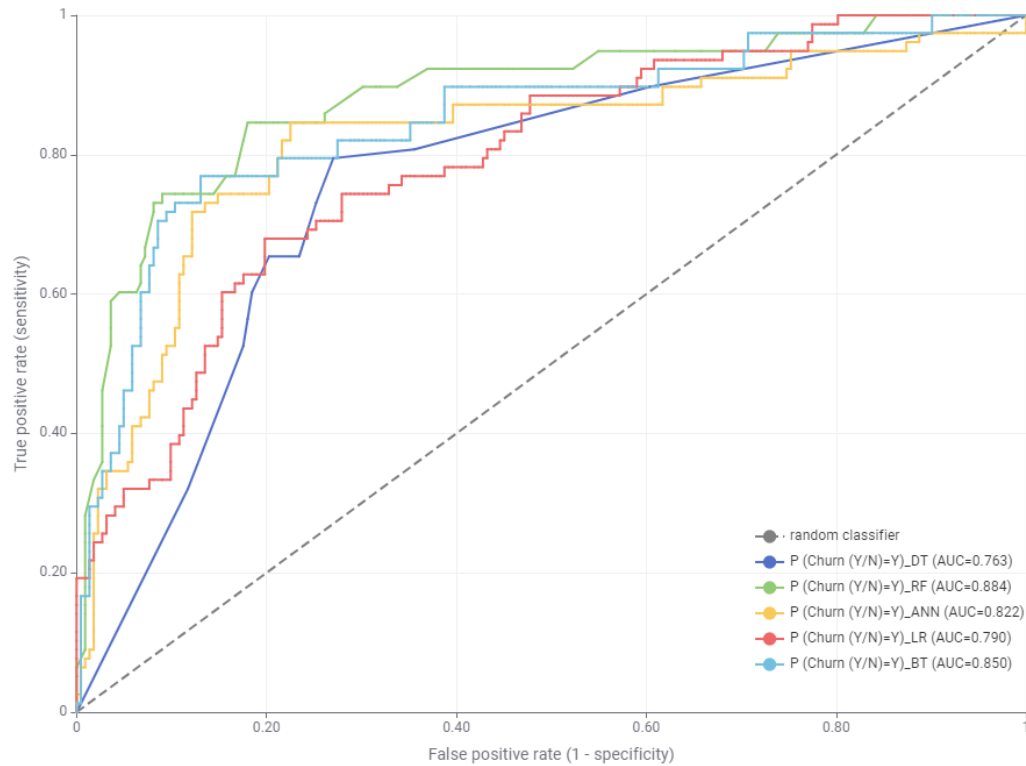


Figure 16 shows the ROC curves for all five models, distinguished by the suffixes in the legend: DT for Decision Tree, RF for Random Forest, ANN for Artificial Neural Network, LR for Logistic Regression, and BT for Boosted Tree. The AUC scores in the bottom right show that the random forest model yielded the highest AUC, indicating that the random forest model is the best at distinguishing between the positive and negative outcomes of customer churn.

Figure 17 – Model Performance Metrics

Model	Sensitivity+N	Sensitivity+Y	Specificity+N	Specificity+Y	Accuracy+O...	Area Under ...
[T] String	[.00] Number (Float)	[.00] Number (Float)	[.00] Number (Float)	[.00] Number (Float)	[.00] Number (Float)	[.00] Number (Float)
Decision Tree	0.796	0.853	0.853	0.796	0.811	0.763
Random Forest	0.829	0.795	0.795	0.829	0.82	0.884
Artificial Neural Ne	0.793	0.795	0.795	0.793	0.793	0.818
Logistic Regression	0.802	0.654	0.654	0.802	0.763	0.79
Boosted Tree	0.824	0.769	0.769	0.824	0.81	0.85

Figure 17 shows the sensitivity and specificity values for the Churn=Y and Churn=N outcomes, as well as the overall accuracy and AUC values of each model. Although the decision tree model seems to have slightly better sensitivity and specificity than the random forest model, the random

forest model has the best overall accuracy in addition to the highest AUC value, making it the best performing model from this report's analysis. I would conclude that the random forest model performs well enough for deployment: the high sensitivity and specificity values indicate that the model does well in identifying true positives and true negatives, the high overall accuracy indicates that the model does well in predicting the correct outcome, and the high AUC value indicates that the model does well in distinguishing the positive and negative outcomes for customer churn.

Figure 18 – Variable Importance Graph

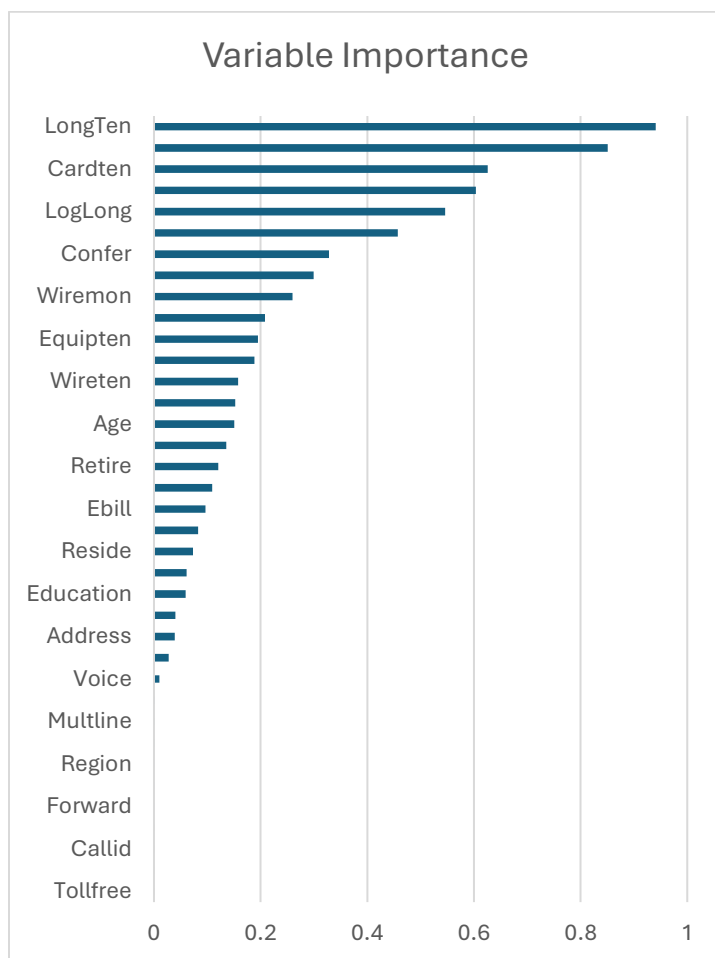


Figure 18 shows the relative importance of each of the predictors in predicting the churn outcome. The graph identifies LongTen as the variable that most contributes to predicting the churn outcome; this aligns with the result of the decision tree in Figure 5, where LongTen was the variable on which the decision tree first split.

Deployment

The model serves several purposes. Firstly, the random forest model can be used to predict which customers are likely to churn, relying on its high overall accuracy and AUC value. Secondly, the model can be analyzed to assess which variables most affect customer churn; these variables, such as the length of a customer's tenure, can be targeted with marketing campaigns to manipulate the values or outcomes of these variables so that the customer would be less likely to churn. And lastly, the predictions of this model can be used to develop targeted marketing strategies to advertise specifically to the customers that are at a higher risk of churning, to hopefully prevent these customers from churning.

Summary/Takeaways

This report simultaneously demonstrates the power of predictive modeling as well as the importance of carefully selecting the best performing model. Although most of the models demonstrated acceptable performance, one clearly outperformed the others. Personally, I learned the importance of selecting the best performing model, since in a professional environment, there is no excuse for choosing the second or third best performing option when a better performing option exists for predictive modeling.

Workflow (Complete Picture)

